Sprint 1

This Sprint was a little slower, but I started it off by giving my team a pitch on where we should take the project. We ended up settling on making it into a kart racer similar to Mario Kart with items and a track designed like a horse head (similar to the design of Yoshi Circuit in Mario Kart Double Dash). At the start of this sprint, my job was to determine the final design for our controller and how to handle steering, as well as how we were going to allow the player to jump.

Situation: Initially, for jumping, we used a sewing machine foot pedal, but decided it was too hard to consistently press, especially while rocking,

Task: I spent a lot of time researching what else we could use when some people on the team recommended using a pressure sensor to simulate stomping. The pressure sensor was flat and was way easier to hit consistently.

Action: So we went with a force pressure sensor instead that we got from a previous team's prototype. I reverse-engineered some code to calculate the force input onto the sensor, and if it was greater than a certain threshold, I would output a 1 to essentially mimic a button click.

Result: The Pressure Sensor is basically acting as a flat button and can easily be adjusted to detect a greater threshold of force if it is too sensitive. I will be hooking it up to the prototype controller next week for further testing.

For the steering, I just grabbed the turn angle from the Arduino kit accelerometer, and with the help of software engineering code, we were able to turn the data into usable steering in the engine. We need the final controller to playtest the accuracy and may pivot to a different sensor, like a rotary sensor, if the accelerometer proves to be

too inaccurate. We are also considering Bluetooth to avoid hardwiring the control, as well as other options for components to make the game feel responsive and the data as accurate as possible.